

Themenbereich: Parameter 12.6

a) $xy' = y$

$$y' = \frac{dy}{dx}$$

$$x \cdot \frac{dy}{dx} = y \quad | \cdot dx$$

$$\frac{x \cdot dy}{x \cdot y} = \frac{y \cdot dx}{x \cdot y} \quad | : y$$

$$\frac{dy}{y} = \frac{dx}{x}$$

$$\ln y = \ln x + \ln C$$

$$\ln y = \ln x \cdot C$$

$$y = x \cdot C$$

b) $\frac{dy}{e^x} = \frac{dx}{y-1}$

$$(y-1) dy = e^x dx$$

$$\int (y-1) dy = \int e^x dx$$

$$\frac{y^2}{2} - y = e^x + C$$

$$2) y' = \frac{3y+5}{x-4}$$

$$\frac{dy}{dx} = \frac{3y+5}{x-4}$$

$$\frac{dy}{3y+5} = \frac{dx}{x-4}$$

$$\int \frac{dy}{3y+5} = \int \frac{dx}{x-4}$$

$$\frac{1}{3} \ln(3y+5) = \ln(x-4) + \ln C$$

$$\ln(3y+5) = 3 \ln(x-4) + \ln C$$

$$\ln(3y+5) = \ln C (x-4)^3$$

$$y = \frac{C(x-4)^3 - 5}{3}$$

15.2

$$a) (x+3)y' = 4y$$

$$(x+3) \frac{dy}{dx} = 4y$$

$$\frac{dy}{y} = \frac{4}{x+3} dx$$

$$y(1) = 3$$

$$\ln y = 4 \ln x + 3 + \ln C$$

$$y = C (x+3)^4$$

$$3 = C (1+3)^4 = C \cdot 256$$

$$y(1) = 3$$

$$y = \frac{3}{256} (x+3)^4$$

$$\int \frac{dy}{x^3+7} = \frac{dx}{y}$$

$$y(0) = 4$$

$$\frac{dy}{x^3+7} = \frac{dx}{y}$$

$$y dy = (x^3+7) dx$$

$$\frac{y^2}{2} = \frac{x^4}{4} + 7x + C$$

$$\frac{16}{2} = 0 + 0 + C$$

$$C = 8$$

$$y(0) = 4$$